Ivan Evdokimov, Ph.D.

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Professional Summary

Software Engineer with 5+ years of production experience building backend systems, cloud infrastructure, and ML pipelines. Delivers across the full stack — PostgreSQL-backed APIs, CI/CD automation, AWS-based MLOps — with 95%+ model accuracy on sensitive datasets.

Experience

Software Engineer – Full Stack & MLOps, University of Essex — Colchester, UK — Nov 2025 — Present

- Built GitHub Actions/Vercel CI/CD pipelines with automated unit and integration testing and code review, supporting an AI-assisted development workflow and enabling weekly releases across a multi-team platform.
- Deployed Dockerised end-to-end MLOps infrastructure for a multi-team platform, reducing deployment time and enabling continuous production monitoring.
- Designed REST APIs using FastAPI to create maintanable and scalable backend systems with LLM integrations.
- Developed a Java mobile app with OpenAI API integration for 20 field technicians, automating inspection reports and saving 3 weeks/year of manual data collection.

Software Engineer – Backend & ML, UK Data Service — Colchester, UK — Jan 2023 — Oct 2025

- Trained, fine-tuned, and deployed domain-specific PyTorch, scikit-learn, LLaMA models using Hugging Face, vLLM, and Ollama for automated metadata classification and disclosure risk assessment on a public data portal, achieving 95%+ F1 accuracy on sensitive social science datasets.
- Deployed and orchestrated ML retraining pipelines using AWS Lambda, Step Functions and infrastructure-as-code practices, with monitoring for drift and accuracy to trigger automated retrains — reducing manual intervention by 50%.
- Built ETL pipelines to extract, transform, and query PostgreSQL-hosted social science datasets of 100K+ records as model inputs for automated metadata classification, improving throughput and classification accuracy at scale.
- Optimized statistical disclosure control algorithms by migrating from R to C++ using bitmask techniques, achieving 11x performance improvement and enabling real-time processing of datasets with 500k+ records.
- Delivered MCP servers, developed a LangGraph-based agent-to-agent (A2A) workflow with hallucination checks to automate exploratory data analysis (EDA) — including dataset loading, statistical description, and property derivation — reducing the time independent researchers spent on manual data analysis.

Research Officer & ML Instructor, University of Essex — Colchester, UK — Oct 2021 — Dec 2022

- Migrated quantitative macroeconomic models from MATLAB to C++, reducing simulation runtime 8x for collaborations with other universities in the UK and the USA.
- Delivered lectures and labs in Data Structures, C/C++, and Machine Learning to 100+ undergrad and postgrad students, achieving an average student feedback score of 4.8/5.

Projects

Chat-LLM — github.com/ivan020/ChatLLM

- Built a self-hosted RAG pipeline over a Raspberry Pi-hosted Ollama LLM using LangChain, pgvector, and all-MiniLM-L6-v2 embeddings, indexing past stream transcripts to allow new viewers query past content in real time via Twitch chat.
- Optimised the TinyLlama-1.1B-Chat-v1.0 model for low-latency inference under constrained hardware.

FinBot: Chat-Based Trading Simulation — github.com/ivan020/FinBot

- Built a Dockerised real-time portfolio tracker in Golang with buy/sell mechanics, a PostgreSQL (Supabase) backend recording all transactions, and a live leaderboard — using Twitch chat as the user interface.
- Developed a TypeScript React frontend with real-time portfolio visualisation, interactive performance graphs, and transaction history, serving 120+ monthly users.

MTG Card Reader — ivan020.github.io/MtgRecognizer/

- Built a Dockerised Golang REST API that uses OCR to extract text from Magic: The Gathering card photos, identifies the card via LLM, and returns real-time price data from a PostgreSQL-backed catalogue.
- Developed a TypeScript React frontend with card filtering, deck management, and real-time price display, deployed on Vercel.

Education

University of Essex, PhD in Computational Finance — Oct 2021 — May 2025

- Developed a PyTorch transfer learning framework for Bayesian model averaging in financial forecasting, first-authored a peer-reviewed CS/ML journal publication, and co-authored 3 international conference papers; released an open-source package with external adoption.

University of Essex, MSc in Financial Econometrics — Oct 2020 — Sep 2021

- Sole-authored a modular C++ agent-based simulation modelling household consumption, employment, and borrowing behaviour under dynamic and negative interest rates, with an extensible architecture supporting additional firm and household modules.